

BiasAperture

Literature Review

1. Introduction

This literature review supports the design of BiasAperture, a reusable software platform for auditing facial-analysis models for demographic bias.

The high-level research question is: how can facial-analysis systems be audited for subgroup and intersectional bias in a way that is repeatable, statistically grounded, explainable, and traceable to concrete regulatory obligations, rather than relying on a one-off manual study?

This problem is broken down into the following sub-problems, each of which is addressed by a distinct strand of the literature reviewed in this chapter:

- What fairness metrics should be computed, and how should they be defined so that different, non-interchangeable notions of fairness (error-rate parity vs outcome-rate parity) are not conflated with one another.
- How a measured disparity should be judged - as a bare threshold comparison, or as a statistically characterized finding with a significance test and confidence interval.
- How to ensure that a detected disparity reflects a genuine demographic effect rather than an unrelated visual confound the model has learned as a proxy.
- What benchmark datasets are suitable for subgroup validation, and how their construction and known limitations should be documented.
- How the platform's output should be structured so that it is legible to both technical and non-technical stakeholders, and how it should be evidenced against a concrete regulatory-technical standard (the EU AI Act).

Chapter 1 established that a mature body of fairness research has not yet converged into a reusable auditing platform. This chapter reviews each contributing work in turn, covering what it studied, what was learned from it, and how this project adopts, extends, or departs from it.

2. General Review

Major Ideas

- Subgroup and intersectional stratification of error rates, rather than a single aggregate accuracy figure, as the empirical basis for detecting bias (Buolamwini & Gebru).
- Formal, error-rate-based fairness definitions - equalized odds and equal opportunity - as genuinely distinct fairness questions rather than interchangeable variants of one another (Hardt et al.).
- Statistical characterisation of a disparity (significance testing and confidence intervals) as an alternative to a bare pass/fail threshold such as the four-fifths rule (Watkins et al.).
- Explainability / feature attribution as a required step to distinguish a genuine demographic effect from a proxy-variable confound (Kurian et al.).

- Standardised, structured reporting for both models (Model Cards) and datasets (Datasheets) as the documentation convention the field already recognises (Mitchell et al.; Gebru et al.).
- Mapping technical verification activities directly to regulatory clauses (e.g., the EU AI Act) for auditable, traceable compliance evidence (Buscemi et al.).

Datasets Available

- FairFace - balanced across a seven-category race taxonomy, gender, and age; adopted as the principal validation benchmark.
- UTKFace - used as a secondary benchmark; its age labels are model-estimated rather than human-annotated, a known source of label noise.
- Pilot Parliaments Benchmark (PPB) - built by Buolamwini & Gebru for balanced skin-type and gender representation, used in the original Gender Shades audit.

Common Models Used for the Problem

- Commercial gender/facial-classification APIs, evaluated as black-box services (Gender Shades).
- Convolutional neural networks (CNNs) trained for facial-attribute and medical-image classification tasks, which can implicitly encode demographic proxies (Kurian et al.; FairFace baselines).

Major Issues with the Problem / Models / Datasets

- Intersectional subgroups can show far larger error-rate gaps than either demographic axis considered alone.
- A single aggregate accuracy figure can mask large subgroup disparities.
- Threshold-based fairness tests (e.g., the four-fifths rule) can create a false sense of legal and ethical safety.
- Models can encode demographic information through unrelated visual features (texture, lighting), so a detected disparity may be a confound rather than a genuine demographic effect.
- Dataset subgroup sample sizes are frequently skewed by construction, undermining the trustworthiness of subgroup comparisons; model-estimated labels (e.g., UTKFace age) add further label noise.
- There is no standardised, reusable workflow for repeating a fairness audit against an arbitrary model; the literature is comparatively rich in metrics and mitigation techniques but comparatively sparse on standardised auditing pipelines.

Reference List Maintained for This Review

- [1] Buolamwini, J. & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. PMLR.
- [2] Dehdashtian, S. et al. (2024). Fairness and Bias Mitigation in Computer Vision: A Survey.
- [5] Mitchell, M. et al. (2019). Model Cards for Model Reporting. FAT*.
- [6] Gebru, T. et al. (2018). Datasheets for Datasets.

- [7] Karkkainen, K. & Joo, J. (2021). FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age. WACV.
- [8] Hardt, M., Price, E. & Srebro, N. (2016). Equality of Opportunity in Supervised Learning. NeurIPS.
- [9] Watkins, E. A. et al. (2022). The Four-Fifths Rule is Not Disparate Impact.
- [10] Kurian, N. C. et al. (2024). Where, Why, and How is Bias Learned in Medical Image Analysis Models?
- [11] Buscemi, A. et al. (2025). Assessing High-Risk AI Systems Under the EU AI Act.

3. Paper Review

Each of the following nine papers is reviewed against the fields requested for the literature review report: main idea and contribution, datasets and models used, code availability, procedure, techniques applied, performance metrics achieved, and the key learning taken from the paper and adopted into this project.

2.1 Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification

- **Paper Title, Author, Date:** Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification - Buolamwini, J. & Gebru, T. (2018)
- **Venue / Link:** Proceedings of Machine Learning Research (PMLR), Conference on Fairness, Accountability and Transparency. Publicly available via the PMLR proceedings archive.
- **Main Idea:** The authors audit three commercial gender-classification services and show that performance is markedly worse for darker-skinned female subjects than for lighter-skinned male subjects, with intersectional error rates as high as 34.7% for the worst-performing subgroup against as low as 0.8% for the best-performing one. The paper makes the empirical case for facial-analysis auditing and introduces the methodological template of stratifying error rates by demographic subgroup and by demographic intersection rather than reporting a single aggregate accuracy figure.
- **Dataset(s) Used:** Pilot Parliaments Benchmark (PPB) - a face dataset the authors built for balanced representation across skin type (Fitzpatrick scale) and gender.
- **Model(s) Used:** Three commercial gender-classification APIs (Microsoft, IBM, Face++) - evaluated as black-box services rather than models trained by the authors.
- **Code Base:** Not published as an open-source toolkit; the audit was a one-off manual study rather than reusable software.
- **Procedure / Configuration / Test:** Manual, point-in-time external audit: each commercial classifier was queried against PPB images and outputs were compared to ground-truth gender and Fitzpatrick skin-type labels, subdivided into four intersectional subgroups.
- **Particular Techniques / Tricks Applied:** Subgroup and intersectional stratification of error rates; use of the Fitzpatrick skin-type scale as a demographic axis.
- **Performance Metrics / Results:** Classification accuracy / error rate, reported per skin type, per gender, and per intersectional subgroup (up to 34.7% error for the worst subgroup vs 0.8% for the best).

- **Key Learning / Finding:** Both the empirical case for facial-analysis auditing and the methodological template of stratified, intersectional error reporting were adopted for this project. Where Gender Shades is a single, manual, point-in-time audit of three commercial services, this project is designed to be reusable software that can point at an arbitrary facial-analysis model and repeat the same subgroup and intersectional comparison automatically, with a significance test that Gender Shades' original methodology did not itself formalise.

2.2 Fairness and Bias Mitigation in Computer Vision: A Survey

- **Paper Title, Author, Date:** Fairness and Bias Mitigation in Computer Vision: A Survey - Dehdashtian, S. et al. (2024)
- **Venue / Link:** arXiv preprint survey. Publicly available preprint.
- **Main Idea:** The authors survey the broader computer-vision fairness literature, cataloguing where bias enters the pipeline - data collection, annotation, training, and deployment - and the families of fairness metrics and mitigation techniques the field has proposed at each stage.
- **Dataset(s) Used:** Not applicable - this is a literature survey rather than an empirical study; it references datasets used across the surveyed works.
- **Model(s) Used:** Not applicable - survey paper; no single model is trained or evaluated.
- **Code Base:** Not applicable.
- **Procedure / Configuration / Test:** Systematic literature survey and taxonomy construction across the CV-fairness field.
- **Particular Techniques / Tricks Applied:** Taxonomic categorisation of bias sources and of fairness-metric / mitigation-technique families.
- **Performance Metrics / Results:** Not applicable - the paper catalogues metrics used elsewhere rather than reporting its own performance results.
- **Key Learning / Finding:** This survey shows the shape of the gap this project fills: a rich taxonomy of what to measure and how to mitigate bias already exists, but comparatively little of the literature standardises how to run that measurement repeatedly as a workflow. Rather than proposing another mitigation technique or another metric, this project sits deliberately on the diagnostic side of the taxonomy, building the reusable auditing pipeline the survey shows is largely absent, while explicitly declining to fold mitigation into the same platform.

2.3 Equality of Opportunity in Supervised Learning

- **Paper Title, Author, Date:** Equality of Opportunity in Supervised Learning - Hardt, M., Price, E. & Srebro, N. (2016)
- **Venue / Link:** Advances in Neural Information Processing Systems (NeurIPS). Publicly available via NeurIPS proceedings.
- **Main Idea:** The authors formally define equalized odds and equal opportunity, two error-rate-based fairness criteria for supervised classifiers. Equalized odds requires parity in both the true-positive rate and the false-positive rate across demographic groups, while equal opportunity relaxes this to parity in the true-positive rate for the positive outcome class only.
- **Dataset(s) Used:** Not tied to a specific dataset - the paper is a theoretical / methodological contribution illustrated with generic classifier examples.

- **Model(s) Used:** Generic binary/ multi-class supervised classifiers used to illustrate the fairness criteria; no single model architecture is central to the contribution.
- **Code Base:** Not applicable in the source material.
- **Procedure / Configuration / Test:** Formal derivation of fairness criteria and post-processing method for achieving equalized odds / equal opportunity from an existing classifier's scores.
- **Particular Techniques / Tricks Applied:** Post-hoc threshold adjustment per group to satisfy the defined fairness criteria.
- **Performance Metrics / Results:** True-positive rate (TPR) and false-positive rate (FPR) parity across demographic groups.
- **Key Learning / Finding:** Equalized odds and equal opportunity are two genuinely different fairness questions rather than one being a stricter version of an otherwise-interchangeable other, so neither is a general substitute for the other. Both definitions were adopted directly as two of this project's Core Four metrics, computed side by side rather than picking one, and paired with DPD and DIR to supply the outcome-rate and ratio-based perspectives that Hardt et al.'s error-rate framing does not itself cover.

2.4 The Four-Fifths Rule is Not Disparate Impact

- **Paper Title, Author, Date:** The Four-Fifths Rule is Not Disparate Impact - Watkins, E. A. et al. (2022)
- **Venue / Link:** arXiv preprint. Publicly available preprint.
- **Main Idea:** The authors critique the U.S. four-fifths rule as it has been imported into algorithmic fairness practice, arguing that treating an 80% disparate-impact threshold borrowed from employment-discrimination law as a binary pass/fail test is an instance of epistemic trespassing that can grant a false sense of legal and ethical safety while conflating bare compliance with actual fairness.
- **Dataset(s) Used:** Not applicable - a conceptual / legal-epistemological critique rather than an empirical study.
- **Model(s) Used:** Not applicable.
- **Code Base:** Not applicable.
- **Procedure / Configuration / Test:** Conceptual and legal analysis of how the four-fifths rule is used (and misused) as an algorithmic fairness threshold.
- **Particular Techniques / Tricks Applied:** Critical / epistemological argumentation rather than a computational technique.
- **Performance Metrics / Results:** Discusses the Disparate Impact Ratio (DIR) as a ratio-based measure, not as its own contributed metric.
- **Key Learning / Finding:** This paper is the direct reason this project's platform never reports a disparity as a ratio against an implicit cut-off. The naive-threshold practice the paper criticises is addressed by pairing every reported DIR value, and every other Core Four metric, with a chi-squared test of independence and a bootstrap confidence interval, so that a disparity is characterised by its statistical strength rather than measured against a borrowed or contested threshold.

2.5 Where, Why, and How is Bias Learned in Medical Image Analysis Models?

- **Paper Title, Author, Date:** Where, Why, and How is Bias Learned in Medical Image Analysis Models? - Kurian, N. C. et al. (2024)
- **Venue / Link:** Peer-reviewed medical-imaging / ML venue. Publicly available via publisher / preprint archive.
- **Main Idea:** The authors study how convolutional networks trained on medical images can learn to encode demographic information as a proxy through unrelated, seemingly neutral visual features, such as texture or lighting, even when no demographic label appears anywhere in the training data.
- **Dataset(s) Used:** Medical imaging datasets with demographic metadata (e.g., patient age, sex, race) used to probe what CNNs implicitly encode.
- **Model(s) Used:** Convolutional neural networks (CNNs) trained for medical image analysis tasks, probed for demographic-proxy encoding.
- **Code Base:** Not specified in the source material.
- **Procedure / Configuration / Test:** Probing / feature-attribution analysis of trained CNNs to identify which visual features correlate with demographic attributes.
- **Particular Techniques / Tricks Applied:** Feature-attribution / proxy-variable analysis of learned representations.
- **Performance Metrics / Results:** Correlation between learned visual features and demographic labels; not a single headline accuracy figure.
- **Key Learning / Finding:** Applied to facial analysis, this finding means a detected subgroup disparity cannot be assumed to arise only from the demographic label an audit is stratifying on; it may instead be an artefact of a correlated visual feature the model has entangled with that label. This project adopts the finding directly by requiring SHAP-based feature attribution on every flagged disparity, so the explainability layer can help distinguish a genuinely demographic effect from an unrelated confound before a finding is reported as evidence.

2.6 Model Cards for Model Reporting

- **Paper Title, Author, Date:** Model Cards for Model Reporting - Mitchell, M. et al. (2019)
- **Venue / Link:** Conference on Fairness, Accountability, and Transparency (FAT*). Publicly available via ACM Digital Library / arXiv.
- **Main Idea:** The authors propose Model Cards, a standardised, structured summary format for a model's intended use, performance characteristics, and evaluation results, aimed at making that information legible to both technical and non-technical stakeholders.
- **Dataset(s) Used:** Not applicable - proposes a documentation format rather than using a specific dataset.
- **Model(s) Used:** Not applicable - the contribution is a reporting template applicable to any model.
- **Code Base:** Model Card template / examples released alongside the paper.
- **Procedure / Configuration / Test:** Design of a structured documentation template with defined sections (intended use, factors, metrics, evaluation data, ethical considerations, caveats).
- **Particular Techniques / Tricks Applied:** Standardised structured-documentation convention.

- **Performance Metrics / Results:** Not applicable - defines what should be reported rather than reporting model performance itself.
- **Key Learning / Finding:** This paper supplies a documentation convention the field already recognises, so this project does not need to invent a report format from scratch. Model Cards is adopted directly as the structural basis for the model-performance side of the generated fairness report, so that a reader familiar with Model Cards from elsewhere does not have to learn an ad hoc format specific to this platform.

2.7 Datasheets for Datasets

- **Paper Title, Author, Date:** Datasheets for Datasets - Gebru, T. et al. (2018)
- **Venue / Link:** Communications of the ACM (originally arXiv preprint). Publicly available via arXiv / ACM Digital Library.
- **Main Idea:** The authors propose Datasheets for Datasets, a complementary convention to Model Cards, documenting a dataset's motivation, composition, collection process, and recommended uses so that downstream users can reason about its limitations.
- **Dataset(s) Used:** Not applicable - proposes a documentation format applicable to any dataset.
- **Model(s) Used:** Not applicable.
- **Code Base:** Datasheet template released alongside the paper.
- **Procedure / Configuration / Test:** Design of a structured dataset-documentation template covering motivation, composition, collection process, preprocessing, uses, distribution, and maintenance.
- **Particular Techniques / Tricks Applied:** Standardised structured-documentation convention (dataset-side counterpart to Model Cards).
- **Performance Metrics / Results:** Not applicable.
- **Key Learning / Finding:** This paper supplies the dataset-side counterpart to Model Cards that this project's report structure needed. The Datasheets convention is adopted for the dataset-provenance section of the generated report, so that FairFace's and UTKFace's collection processes and known limitations, such as UTKFace's model-estimated age labels, are documented in a recognised structure rather than an ad hoc one.

2.8 FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age

- **Paper Title, Author, Date:** FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age - Karkkainen, K. & Joo, J. (2021)
- **Venue / Link:** IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). Publicly available via IEEE Xplore / arXiv; dataset available on GitHub.
- **Main Idea:** The authors construct FairFace, a face-attribute dataset explicitly built for balanced representation across race, gender, and age, using a seven-category race taxonomy designed to reduce the classification skew present in earlier face datasets.
- **Dataset(s) Used:** FairFace - a balanced, human-annotated face dataset spanning seven race categories, gender, and age groups.
- **Model(s) Used:** Baseline face-attribute classification models (e.g., CNN-based classifiers) trained/evaluated on FairFace by the original authors.
- **Code Base:** Dataset and baseline training code publicly released on GitHub.

- **Procedure / Configuration / Test:** Dataset construction with explicit balancing across race/gender/age, followed by baseline classifier training and cross-dataset generalisation evaluation.
- **Particular Techniques / Tricks Applied:** Balanced-sampling dataset construction; cross-dataset generalisation testing.
- **Performance Metrics / Results:** Classification accuracy per attribute (race, gender, age), and generalisation gap across datasets.
- **Key Learning / Finding:** FairFace's balanced construction makes it a more trustworthy basis for subgroup comparison than a dataset whose subgroup sample sizes are skewed by construction. This project adopts FairFace as its principal validation benchmark for exactly this reason, and uses UTKFace only as a secondary benchmark, since UTKFace's age labels are model-estimated rather than human-annotated, a source of label noise the platform's sample-size and confidence-interval reporting is designed to make transparent rather than obscure.

2.9 Assessing High-Risk AI Systems Under the EU AI Act

- **Paper Title, Author, Date:** Assessing High-Risk AI Systems Under the EU AI Act - Buscemi, A. et al. (2025)
- **Venue / Link:** Publicly available preprint / conference paper. Publicly available preprint.
- **Main Idea:** The authors decompose the EU AI Act's high-risk-system obligations into a stable taxonomy of high-level requirements and sub-requirements, mapping each to a concrete, technology-agnostic technical verification activity.
- **Dataset(s) Used:** Not applicable - a regulatory/legal-technical mapping paper rather than an empirical study.
- **Model(s) Used:** Not applicable.
- **Code Base:** Not applicable.
- **Procedure / Configuration / Test:** Decomposition of EU AI Act Article 10-type obligations into checkable, technology-agnostic verification activities.
- **Particular Techniques / Tricks Applied:** Regulatory-requirement-to-technical-activity mapping / taxonomy construction.
- **Performance Metrics / Results:** Not applicable.
- **Key Learning / Finding:** This paper shows how to turn a legal article into a specific, checkable test rather than a general statement of intent. This project follows the same operational spirit at the scale of a single Article 10 mapping: Section 4.5 ties each Core Four metric, each significance test, and each sample-size guard to the specific data-governance clause it evidences, rather than the generated report gesturing at compliance in general terms.

4. Literature Review Matrix

The matrix below summarises the nine papers reviewed against the standard Literature Review Matrix fields (Walden University Academic Guides format).

Title/Author/Date	Conceptual Framework	Research Question(s)/Hypotheses	Datasets	Methodology	Analysis & Results	Conclusions	Implications for Future Research
Gender Shades (Buolamwini & Gebu, 2018)	Intersectional / subgroup auditing of commercial classifiers	Do commercial gender-classification services perform equally across skin type and gender intersections?	Pilot Parliaments Benchmark (PPB)	Manual point-in-time external audit of 3 commercial APIs	Error rates up to 34.7% for worst subgroup vs 0.8% for best	Commercial classifiers show large intersectional accuracy disparities	Motivated building a reusable, automated, repeatable auditing pipeline (this project) instead of a one-off manual audit
Fairness and Bias Mitigation in CV: A Survey (Dehdashtian et al., 2024)	Taxonomy of CV bias sources and fairness-metric / mitigation families	Where does bias enter the computer-vision pipeline and what has the field proposed to address it?	Not applicable (survey)	Systematic literature survey and taxonomy construction	Rich taxonomy of what to measure and how to mitigate; little standardisation of how to run measurement repeatedly	The auditing-workflow literature is immature relative to the metrics literature	Motivated the reusable, diagnostic-only auditing pipeline, deliberately excluding mitigation
Equality of Opportunity in Supervised Learning (Hardt et al., 2016)	Error-rate-based formal fairness definitions	How can equalized error rates across groups be formally defined and achieved?	Not tied to a specific dataset	Formal definition + post-processing threshold adjustment	Formal definitions of equalized odds and equal opportunity established	Equalized odds and equal opportunity are distinct, non-interchangeable fairness criteria	Adopted directly as the EOD and EOP Core Four metrics, computed together rather than one substituting for the other
The Four-Fifths Rule is Not Disparate Impact (Watkins et al., 2022)	Critique of threshold-based legal fairness tests	Is the 80% four-fifths rule a valid binary fairness test when imported from employment law?	Not applicable (conceptual/legal analysis)	Legal / epistemological critique and argumentation	The four-fifths threshold is shown to be borrowed and epistemically ungrounded as a pass/fail test	Bare compliance with the four-fifths rule is not equivalent to actual fairness	Motivated pairing every DIR value and Core Four metric with a chi-squared test and bootstrap confidence interval instead of a bare threshold
Where, Why, and How is Bias Learned in Medical Image Analysis Models? (Kurian et al., 2024)	Proxy-variable / feature-attribution analysis of CNNs	Can CNNs encode demographic information via unrelated visual features?	Medical imaging datasets with demographic metadata	Feature-attribution probing of trained CNNs	CNNs can learn demographic proxies through unrelated features such as texture or lighting	A detected subgroup disparity may be a confound rather than a genuine demographic effect	Motivated requiring SHAP-based explainability on every flagged disparity before it is reported as evidence
Model Cards for Model Reporting (Mitchell et al., 2019)	Standardised model-reporting documentation	How can model performance and evaluation results be reported legibly to diverse stakeholders?	Not applicable	Structured documentation template design	A reusable Model Card template with defined sections is proposed	Standardised reporting improves legibility for technical and non-technical stakeholders	Adopted as the structural basis for the model-performance section of the generated fairness report

Title/Author/Date	Conceptual Framework	Research Question(s)/Hypotheses	Datasets	Methodology	Analysis & Results	Conclusions	Implications for Future Research
Datasheets for Datasets (Gebru et al., 2018)	Standardised dataset-documentation	How can a dataset's provenance, composition, and limitations be documented for downstream users?	Not applicable	Structured documentation template design	A reusable Datasheet template covering motivation, composition, collection, and limitations	Standardised dataset documentation improves downstream reasoning about limitations	Adopted for the dataset-provenance section, documenting FairFace's and UTKFace's collection processes and limitations
FairFace (Karkkainen & Joo, 2021)	Balanced face-attribute dataset construction	Can a face dataset be constructed to balance race, gender, and age representation?	FairFace (7-category race taxonomy)	Balanced dataset construction + baseline classifier evaluation	FairFace reduces classification skew present in earlier face datasets	Balanced construction provides a more trustworthy basis for subgroup comparison	Adopted as the platform's principal validation benchmark; UTKFace used only as a secondary, noisier benchmark
Assessing High-Risk AI Systems Under the EU AI Act (Buscemi et al., 2025)	Regulatory-to-technical requirement mapping	How can EU AI Act high-risk obligations be mapped to concrete technical verification activities?	Not applicable	Decomposition / taxonomy mapping of legal requirements to technical tests	A stable taxonomy linking Article-level obligations to checkable technical activities	Legal obligations can be operationalised as specific, checkable technical tests	Motivated the Article 10-to-metric traceability mapping (Section 4.5) tying each Core Four metric to a specific data-governance clause

5. Summary Table

Table 5-1. Summary of the literature reviewed, highlighting key contributions and motivations for this work.

Authors	Paper / Title (Year)	Key Contribution	Motivation / Adopted Concepts
Buolamwini, J. & Gebru, T.	Gender Shades (2018)	Audits commercial gender classifiers and reveals intersectional accuracy disparities by skin tone and gender.	Established the empirical case and subgroup/intersectional audit methodology; motivated building this into reusable software rather than a one-off audit.
Dehdashtian, S. et al.	Fairness and Bias Mitigation in Computer Vision (2024)	Surveys where bias enters the CV pipeline and catalogues fairness-metric and mitigation families.	Showed the auditing-workflow literature is immature relative to the metrics literature; motivated the reusable, diagnostic-only pipeline.
Hardt, M., Price, E. & Srebro, N.	Equality of Opportunity in Supervised Learning (2016)	Formally defines equalized odds and equal opportunity as distinct fairness criteria for supervised classifiers.	Adopted directly as the EOD and EOP Core Four metrics, computed together rather than one substituting for the other.
Watkins, E. A. et al.	The Four-Fifths Rule is Not Disparate Impact (2022)	Critiques the four-fifths threshold as epistemic trespassing when used as a binary pass/fail fairness test.	Motivated pairing every DIR value and Core Four metric with a chi-squared test and bootstrap confidence interval, instead of a bare threshold.
Kurian, N. C. et al.	Where, Why, and How is Bias Learned in Medical Image Analysis Models? (2024)	Shows CNNs can learn to encode demographic information as a proxy through unrelated visual features.	Motivated the SHAP-based explainability layer, run on every flagged disparity, to separate genuine demographic effects from confounds.
Mitchell, M. et al.	Model Cards for Model Reporting (2019)	Proposes a standardised, stakeholder-legible summary format for model reporting.	Adopted as the structural basis for the model-performance section of the generated fairness report.
Gebru, T. et al.	Datasheets for Datasets (2018)	Proposes a documentation convention for dataset motivation, composition, collection process, and limitations.	Adopted as the structural basis for the dataset-provenance section of the generated report.
Karkkainen, K. & Joo, J.	FairFace (2021)	Constructs FairFace, a face-attribute dataset built for balanced representation across race, gender, and age.	Adopted as the platform's principal validation benchmark; UTKFace used only as a secondary, noisier benchmark.
Buscemi, A. et al.	Assessing High-Risk AI Systems Under the EU AI Act (2025)	Maps EU AI Act high-risk obligations to concrete, technology-agnostic technical verification activities.	Motivated the Article 10-to-metric traceability mapping (Section 4.5) that ties each Core Four metric to a specific data-governance clause.

6. Project Solution Proposal

With everything above considered, BiasAperture combines what none of these works do alone: a reusable software pipeline rather than a one-off manual audit; the Core Four metrics (equalized odds, equal opportunity, demographic parity difference, and disparate impact ratio) computed by cross-validating, independently maintained toolkits rather than a single implementation; every

disparity paired with the significance testing Watkins et al.'s critique shows is necessary rather than a bare threshold; SHAP-based diagnosis of the proxy-variable risk Kurian et al. identify; a report structured against the Model Cards and Datasheets conventions; and explicit, clause-level traceability to the EU AI Act (and, in the operational spirit Buscemi et al. establish, comparable technical-verification frameworks such as the NIST AI Risk Management Framework).

This project's next steps follow the requirement document and architecture design conventions of the guideline: the requirement document is updated with the knowledge gathered in this review, and a tentative architecture diagram for BiasAperture is prepared to specify how the Core Four metrics, the significance-testing layer, the SHAP-based explainability layer, and the Model Cards / Datasheets-structured reporting layer fit together as a single reusable pipeline.

7. References

- [1] Buolamwini, J., & Gebru, T. (2018). Gender Shades: Intersectional accuracy disparities in commercial gender classification. *Proceedings of Machine Learning Research*, 81, 77-91.
- [2] Dehdashtian, S., et al. (2024). Fairness and bias mitigation in computer vision: A survey. *arXiv preprint*.
- [5] Mitchell, M., et al. (2019). Model cards for model reporting. *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*.
- [6] Gebru, T., et al. (2018). Datasheets for datasets. *arXiv preprint / Communications of the ACM*.
- [7] Karkkainen, K., & Joo, J. (2021). FairFace: Face attribute dataset for balanced race, gender, and age. *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.
- [8] Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [9] Watkins, E. A., et al. (2022). The four-fifths rule is not disparate impact. *arXiv preprint*.
- [10] Kurian, N. C., et al. (2024). Where, why, and how is bias learned in medical image analysis models?
- [11] Buscemi, A., et al. (2025). Assessing high-risk AI systems under the EU AI Act.

Literature Review Matrix reference format: Walden University Academic Guides. Retrieved from https://academicguides.waldenu.edu/ld.php?content_id=615424